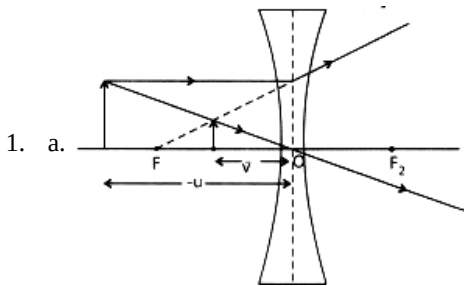


Solution

LIGHT -03

Class 10 - Science



b. The object distance (u) and image distance (v) are marked in the diagram of part (a).

Relation: $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$

c. As, $m = -1$; hence, the lens is convex.

Now, Magnification, $m = \frac{v}{u} = -1$

$\therefore v = -u$

Thus object is at $2f$.

$2f = 40\text{cm}$

$f = 20\text{cm} = 0.2\text{m}$.

$P = \frac{1}{f} = \frac{1}{0.2} = +5\text{D (convex lens)}$.

2. Given,

$f = 30\text{cm}$

$u = -50\text{ cm}$

$h = 6\text{cm}$

Lens formula

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\frac{1}{v} = \frac{1}{f} + \frac{1}{u}$$

$$\frac{1}{v} = \frac{1}{30} - \frac{1}{50}$$

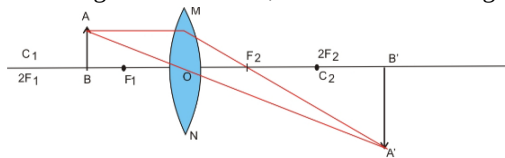
$$v = +75\text{ cm}$$

$$m = \frac{v}{u} = \frac{h_i}{h}$$

$$\frac{75}{-50} = \frac{h_i}{6}$$

$$h_i = -9\text{cm}$$

The image formed is real, increased and enlarged.



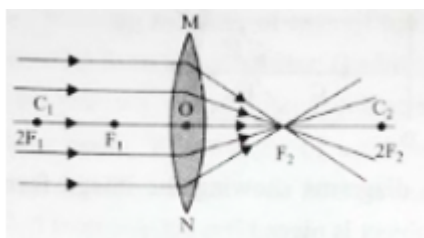
3. The degree of convergence and divergence provided by a lens is called the power of the lens. The unit of power of lens is Diopter D.

The focal length of the lens used by the first student is positive hence it is a convex lens. The lens of the second student is a concave lens.

$$p = \frac{1}{f} = \frac{1}{0.5} = 2 \text{ Power of lens (first student)} = +2$$

$$\text{Power of lens (second student)} = -2$$

4.



5. The refractive index of a medium is given by

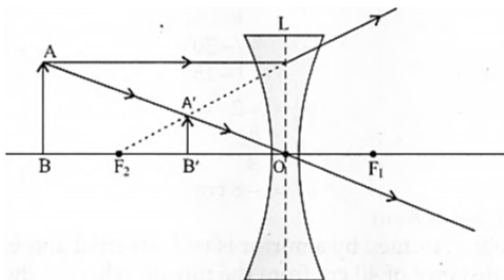
$$n_m = \frac{\text{speed of light in air}}{\text{speed of light in medium}} = \frac{c}{v}$$

Let v_1 and v_2 is the speed of light in medium 1 and medium 2 respectively.

The refractive index of medium 2 with respect to medium 1 is given by the ratio of the speed of light in medium 1 and in medium 2 can be represented as:

$$n_{21} = \frac{v_1}{v_2}$$

6. It is a concave lens.



The power of a lens is given by the relation

$$\therefore P = \frac{1}{f(\text{in metre})}$$

$$P = 10 \text{ D}$$

$$\therefore f = \frac{1}{P} = \frac{1}{10} = 0.1 \text{ m}$$

or -10 cm, as lens is concave lens.

7. The light ray travelling in air and entering obliquely into water is bent towards the normal, because the speed of the light decreases in water (water is optically denser than air).

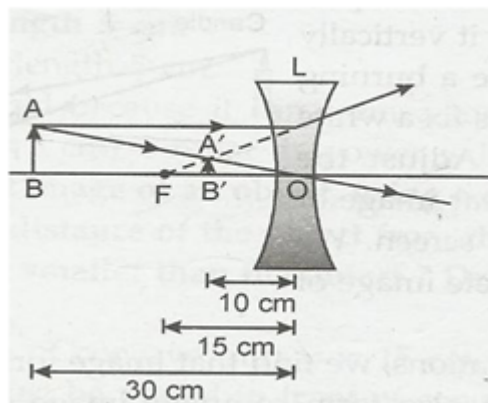
8. $f = -15 \text{ cm}$, $v = -10 \text{ cm}$

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{u} = \frac{1}{15} - \frac{1}{10} = -\frac{1}{30}$$

$$u = -30 \text{ cm}$$

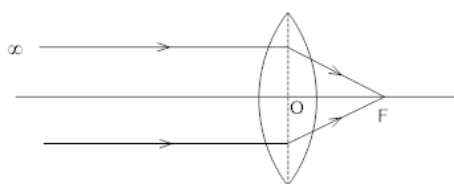
Ray diagram as follows:



9. i. Size – Point sized

Position – At focus

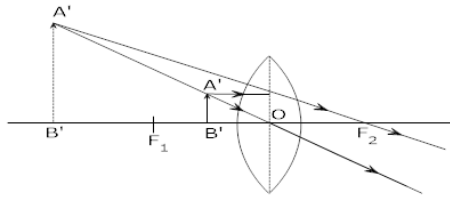
Nature – Real & Inverted



ii. Size – highly magnified

Position – same side of the lens where the object is placed

Nature – virtual & erect.



10. If n number of lenses are placed in close contact, then the power of the combination of lenses is equal to algebraic sum of the powers of the individual lenses.

$$P = P_1 + P_2 + \dots + P_n$$

11. Since the image is real and diminished, the lens must be convex and the object must be placed beyond $2F$.

Given:

$$v = +18 \text{ cm}$$

$$\frac{h'}{h} = \frac{-4}{5} \dots [m = \frac{h'}{h} \text{ is negative for real image}]$$

$$f = ?$$

$$\text{We have, } m = \frac{h'}{h} = \frac{v}{u}$$

$$\frac{v}{u} = \frac{-4}{5}$$

$$\text{and } 4u = -5 \times 18 \text{ cm}$$

$$\text{Therefore, } u = -22.5 \text{ cm}$$

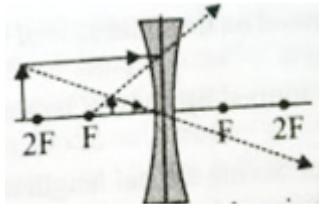
Also focal length is given by

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\frac{1}{18} - \frac{1}{-22.5} = \frac{1}{f} + \frac{1}{22.5}$$

$$= \frac{9}{90}$$

$$\text{Therefore, } f = 10 \text{ cm}$$



13. The steps to be followed are:

- i. Clean the microscope and the lenses.
- ii. Place the slide on the stage.
- iii. Place the low-power objective lens above the slide.
- iv. Focus the material with the help of a coarse adjustment.
- v. Sharpen the focus with fine adjustment.
- vi. Bring high power objective lens over the slide.

14. a. Concave lens.

b. Optical centre.

c. Given: $u = -30 \text{ cm}$, $f = +20 \text{ cm}$

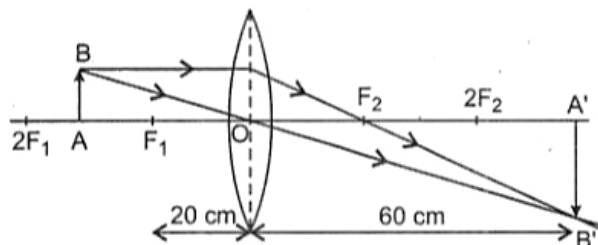
i. Using lens formula,

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\frac{1}{v} = \frac{1}{f} + \frac{1}{u} = \frac{1}{20} + \frac{1}{-30} = \frac{3-2}{60}$$

$$\Rightarrow v = 60 \text{ cm}$$

The image is formed at a distance of 60 cm from the lens on the right side.



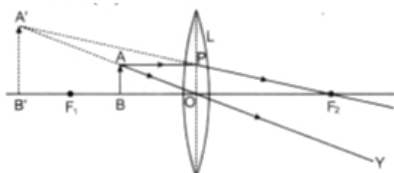
ii. $\therefore m = \frac{v}{u} = \frac{60}{-30} = -2$

So, the image is inverted and double the size of the object.

iii. The image is real as $v > 0$, inverted and enlarged.

15. Here, LN is parallel to AB. Therefore, LN is the correct emergent ray.

16. A convex lens forms a virtual and erect image when the object is placed between the optical centre (O) and focus (F').



17. Refractive index of glass = $\frac{\text{Speed of light in air}}{\text{Speed of light in glass}}$

or, $1.50 = \frac{\text{Speed of light in air}}{\text{Speed of light in glass}}$

or, Speed of light in glass = $\frac{3 \times 10^8}{1.50} \text{ ms}^{-1}$

$= 2 \times 10^8 \text{ ms}^{-1}$

The speed of light in glass is $2 \times 10^8 \text{ ms}^{-1}$.

18. Yes, the half covered lens will still produce a complete image of the object but the image so formed may not be as intense as formed with uncovered lens. In fact, parts or broken pieces of lens behave like a complete lens and form complete image.

Verification : Take a convex lens. Light a candle. Now form an image of a burning candle on a white surface on the other side of lens by adjusting the distance between lens and candle. We can observe, a complete real and an inverted image of candle. Now cover half of lens with black paper, and try to form an image now. We can observe that again a complete, real and an inverted image of candle is formed. Image formed was less intense from earlier.

19. Focal length = $\frac{1}{P} = \frac{1}{4} = 0.25 \text{ m}$ or 25 cm. It is a convex lens as power and focal length are positive.

Focal length, $f = +25 \text{ cm}$

Object distance, $u = -50 \text{ cm}$

Image distance, $v = ?$

Image height, $l = ?$

Using lens formula,

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\frac{1}{25} = \frac{1}{v} - \frac{1}{(-50)}$$

$$v = +50 \text{ cm}$$

Also, magnification, $m = -\frac{v}{u}$

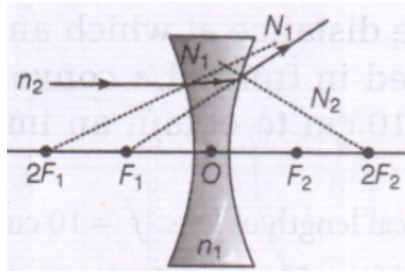
Hence, $m = \frac{(50)}{-(-50)} = -1$

Image formed 50 cm in front of the lens. Due to a magnification of 1, the image is the same size as the object. The image formed is inverted as evidenced by the negative magnification.

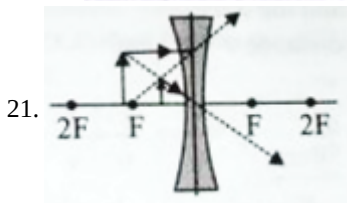
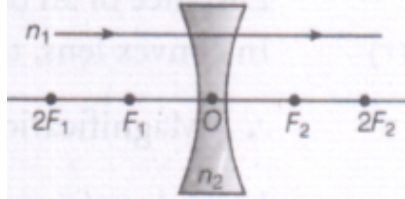
20. i. In this case, $n_1 > n_2$

i.e. medium of the concave lens is denser than the medium from where light ray is incident on it. Therefore, the light bends towards the normal N_1 after entering the glass surface. While leaving the lens, the ray again bends away from the normal as

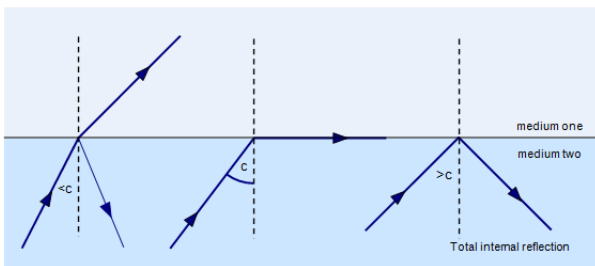
light is going from a denser medium to rarer medium



ii. In this situation when $n_1 = n_2$, light ray goes parallel undeviated as the refractive indices of both media are same.



22. The following figure shows a ray of light incident obliquely. Every medium has a critical angle. When angle of incidence is more than critical angle, the ray of light comes back in the same medium. This phenomenon is called total internal reflection.



23. $u = ?$; $v = -10$ cm [real image]

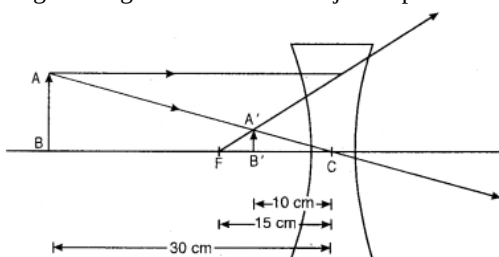
$f = -15$ cm [concave lens];

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u} \text{ or } \frac{1}{u} = \frac{1}{v} - \frac{1}{f}$$

$$\frac{1}{u} = \frac{1}{-10} - \frac{1}{(-15)} = -\frac{1}{10} + \frac{1}{15} = \frac{-3 + 2}{30} = \frac{-1}{30}$$

$$u = -30 \text{ cm}$$

Negative sign shows that the object is placed to the left of lens.



24. 1 dioptre is SI unit of the power of a lens whose focal length is 1 metre. It is denoted by the letter D. Thus $1D = 1 \text{ m}^{-1}$. Simply, when focal length 'f' is expressed in metres, then, power is expressed in dioptres. The power of a convex lens is positive and that of a concave lens is negative.

25. Focal length of convex lens = 20 cm

Object distance, $u = -15$ cm

Using the lens formula,

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

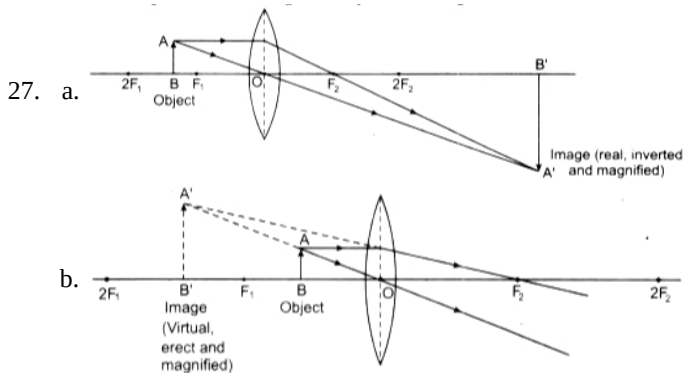
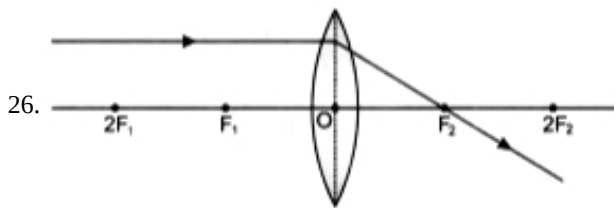
$$\frac{1}{20} = \frac{1}{v} + \frac{1}{15}$$

$$\frac{1}{v} = \frac{1}{20} - \frac{1}{15}$$

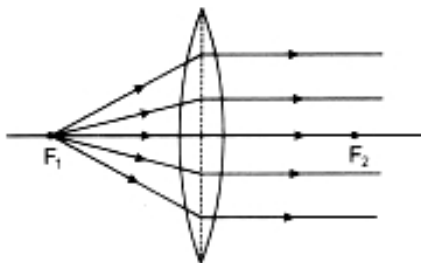
$$v = -60 \text{ cm}$$

Four characteristics of the image formed by the lens are:

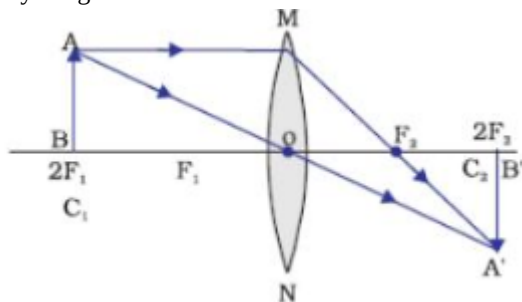
- i. Virtual
- ii. Erect
- iii. At a distance of 60 cm on the same side of the lens as the object.
- iv. Enlarged image.



28. The beam of light coming out of lens is parallel beam of light as shown.



29. a. As the image is real, inverted and of same size as that of object, the lens should be used is Convex lens.
 b. This happened only when an object placed at $2F_1$. Therefore, the focal length of the lens is 2 metre. The image is formed at $2F_2$, therefore the distance of candle flame from the lens is 4 metre.
 c. Ray Diagram is shown below:



30. Power of a lens is its ability to converge or diverge the rays of light falling on it. Power of a lens is equal to reciprocal of the focal length of the lens. SI unit of power is dioptre (D).

$$f = -10\text{cm} = -0.1\text{m}$$

$$p = \frac{1}{f} = \frac{1}{(-0.1)} = -10\text{m}$$

It is a concave lens.

Object distance ,u = -20 cm

Focal length, f = -10 cm (concave lens)

Lens formula

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\frac{1}{v} = \frac{1}{f} + \frac{1}{u}$$

$$\frac{1}{v} = \frac{1}{10} + \frac{1}{-20} = \frac{-3}{20}$$

$$v = -6.66\text{ cm}$$

hence, the image is 6.6 cm in front of the lens on the same side as the object

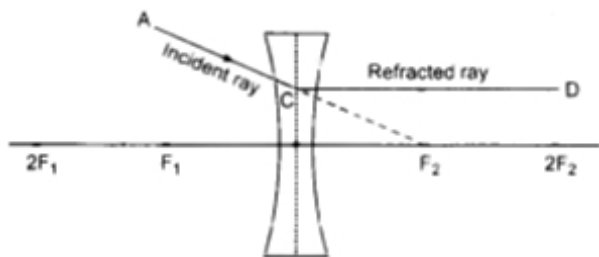
For a lens, magnification is

$$m = \frac{v}{u}$$

$$m = \frac{-6.6}{-20}$$

$$m = +2.34$$

31. In figure the refracted ray parallel to the principal axis. So, the incident ray must be appearing to meet at the principal focus of concave lens. To find the incident ray, F_2 is joined to C and produced as shown in the figure.



32. Focal length of concave lens = - 2 m

$$P = -1/f = 1/(-2m)$$

$$= -0.5 \text{ D}$$

$$33. f = \frac{1}{P} = \frac{1}{1.5} = 0.67 \text{ m}$$

$$f = 0.67 \text{ m}$$

Lens is converging.

34. Given, $P = +5 \text{ D}$

We have,

$$\text{Power, } P = \frac{1}{f(\text{in meter})}$$

$$+5 = \frac{1}{f}$$

$$f = \frac{1}{5} \text{ m}$$

$$= \frac{100}{5} = 20 \text{ cm}$$

Focal length, $f = 20 \text{ cm}$ (or + 20 cm).

Since, focal length of the lens is positive. Therefore, the nature of lens is convex. Same size and inverted image is formed when

Magnification,

$$m = -1$$

$$\text{Also, } m = \frac{v}{u}$$

$$v = -u$$

From the lens formula,

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$-\frac{1}{u} - \frac{1}{u} = \frac{1}{f}$$

$$\frac{-2}{u} = \frac{1}{f}$$

$$u = -2f$$

$$u = -2 \times 20 [\because f = 20 \text{ cm}]$$

$$= -40 \text{ cm}$$

35. i. In the above figure,

$$\angle N'OO' = \angle MO'O = \angle r = \text{angle of refraction}$$

lateral displacement = Distance O'B

- ii. Given, refractive index from air to glass is $\frac{3}{2}$.

$$\text{i.e. } {}_a\mu_g = \frac{\mu_g}{\mu_a} = \frac{3}{2}$$

and we know that,

$$\Rightarrow {}_a\mu_g \times {}_g\mu_a = 1$$

$$\Rightarrow {}_g\mu_a = \frac{1}{{}_a\mu_g}$$

$$\Rightarrow {}_g\mu_a = \frac{1}{\left(\frac{3}{2}\right)} \Rightarrow {}_g\mu_a = \frac{2}{3}$$

i.e. refractive index of air for light going from glass to air is ${}_g\mu_a = \frac{2}{3}$

$$36. n_{ia} = \frac{n_i}{n_a} = 1.31 \dots \dots \dots (i)$$

$$n_{ra} = \frac{n_r}{n_a} = 1.54 \dots \dots \dots (ii)$$

$$n_{ri} = \frac{n_r}{n_i} = \frac{n_r}{\frac{n_a}{n_i}}$$

$$n_{ri} = \frac{n_r}{n_a} \times \frac{n_a}{n_i} = \frac{1.54}{1.31} = 1.175$$

37. Lateral shift depends on the following factors:

- i. It is directly proportional to the thickness of glass slab.
- ii. It is directly proportional to the angle of incidence.
- iii. It is directly proportional to the refractive index of the glass slab.
- iv. It is inversely proportional to the wavelength of the light used.

$$38. \text{Refractive index of a medium} = \frac{\text{Speed of light in air}}{\text{Speed of light in the medium}}$$

$$\text{For glass, } \frac{3}{2} = \frac{\text{Speed of light in air}}{2 \times 10^8}$$

$$\text{Speed of light in air} = \frac{3}{2} \times 2 \times 10^8 = 3 \times 10^8 \text{ m/s}$$

$$\text{For water, } \frac{4}{3} = \frac{\text{Speed of light in air}}{\text{Speed of light in water}}$$

$$\text{Speed of light in water} = 3 \times 10^8 \times \frac{3}{4} = 2.25 \times 10^8 \text{ m/s}$$

39. As per question, refractive index of medium x w.r.t medium y, $n_{xy} = \frac{2}{3}$

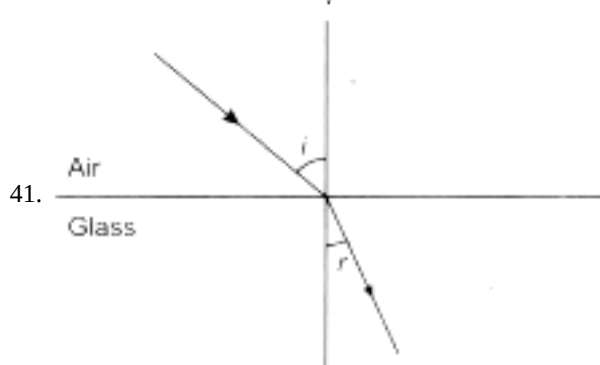
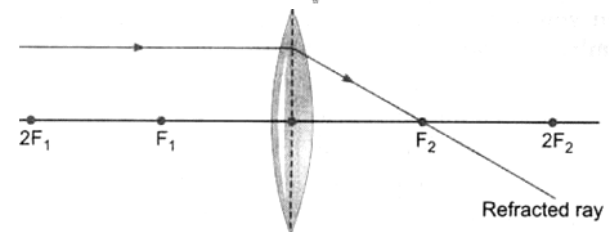
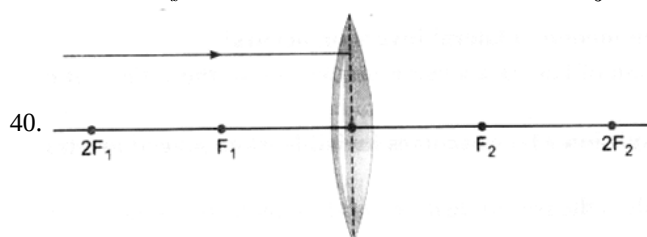
and refractive index of medium y w.r.t. medium z, $n_{yz} = \frac{4}{3}$

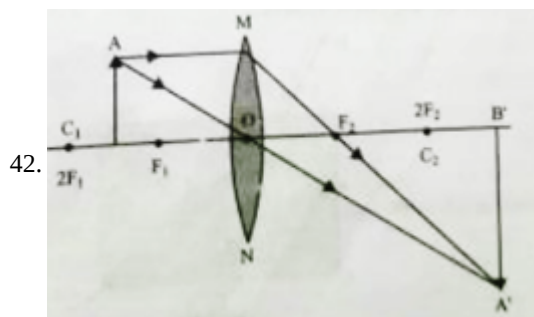
and speed of light in medium x, $V_x = 3 \times 10^8 \text{ ms}^{-1}$

$\therefore$ Refractive index of medium z w.r.t. medium x will be

$$n_{zx} = \frac{n_{zy}}{n_{xy}} = \frac{1}{n_{yz} \times n_{xy}} = \frac{1}{\frac{4}{3} \times \frac{2}{3}} = \frac{9}{8}$$

as, $n_{xy} = \frac{v_y}{v_x}$, Hence $v_y = v_x \times n_{xy} = 3 \times 10^8 \times \frac{2}{3} = 2 \times 10^8 \text{ ms}^{-1}$





43. There are mainly two laws of refraction of light

(i) The incident ray, the normal and the refracted ray, all lie in the same plane.

(ii) If the ray goes from an optically rarer medium to an optically denser medium, it bends towards normal. When the light goes from air to glass, it bends towards the normal. However if the ray goes from an optically denser medium (glass) to an optically rarer medium (air), it bends away from the normal.

(iii) Ratio of sine of angle of incidence to sine of angle of refraction is constant.

$$\text{i.e. } \frac{\sin i}{\sin r} = \mu \text{ constant}$$

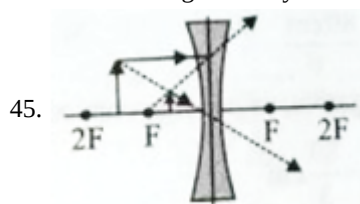
where μ is the refractive index of second medium with respect to first medium.

44.

$$f = \frac{1}{P} = \frac{1}{-2} = -0.5 \text{ m}$$

$$f = -0.5 \text{ m} = -50 \text{ cm}$$

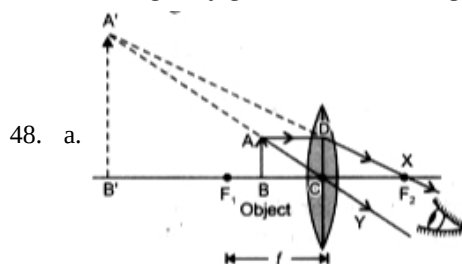
Lens is divergent or say concave.



46. The refractive index for the light going from medium '1' to medium '2' is e to the reciprocal of the refractive index for light going from medium '2' medium '1'. $n_{21} = \frac{1}{n_{12}}$

47. i. Since refracting surfaces in the glass slab are parallel and the ray inside the glass falls at **angle r (angle of refraction)** due to which the emergent emerges at **angle i (angle of incidence)**. Therefore the emergent ray is parallel to the incident ray.

ii. The light ray goes undeviated along the same straight line means **no refraction occurs**.



b. The three characteristics of the image formed are:

- i. Virtual and erect
- ii. magnified
- iii. behind the object.

49.

Real Image	Virtual image
1. The image can be captured on screen .	1.The image cannot be captured on screen .
2.The image is always inverted	2.The image is always erect.
3. Light ray actually meet to form a real image.	3.Light ray do not meet to form a Virtual image

50. i. $n_g = \frac{4}{3}$, $n_w = \frac{3}{2}$, $v_g = 2 \times 10^8 \text{ m/s}$

$$\text{We know that } n_g = \frac{c}{v_g}$$

$$\therefore c = n_g v_g$$

$$= \frac{4}{3} \times 2 \times 10^8 = 2.67 \times 10^8 \text{ m/s}$$

ii. Also, $n_w = \frac{c}{v_w}$
 $\therefore v_w = \frac{c}{n_w} = \frac{2.67 \times 10^8 \times 2}{3}$
 $= 1.78 \times 10^8 \text{ m/s}$

51. i. The lens formula for a concave lens is:

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

Substituting the given values, we get:

$$\frac{1}{-18} = \frac{1}{v} - \frac{1}{-20}$$

Simplifying the above equation, we get:

$$\frac{1}{v} = \frac{1}{-18} - \frac{1}{-20}$$

$$v = -180$$

$v = -180 \text{ cm}$ (negative sign indicates that the image is formed behind the lens)

Therefore, the image distance is 180 cm.

The magnification is given by the formula:

$$m = \frac{-v}{u}$$

Substituting the values, we get:

$$m = \frac{-(-180)}{(-20)}$$

$$m = -9$$

Therefore, the magnification is -9.

- ii. A concave lens always forms a virtual, erect, and diminished image of a real object. On the other hand, when a convex lens forms a virtual image, the magnification can be greater than or less than 1, depending on the position of the object relative to the lens.

52. S.I unit of power is Dioptre (D)

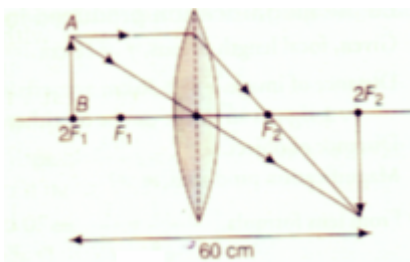
$$F = 50 \text{ cm}$$

$$P = \frac{1}{f}$$

$$P = \frac{1}{50} \times 100$$

$$P = 2 \text{ D}$$

53. i. A ray of light parallel to the principal axis of the concave lens appears to be coming from focus after refraction through the lens.
 ii. A ray of light crossing through a principal focus, after refraction from a convex lens, will appear parallel to the principal axis.
 iii. A ray of light passing through the optical center suffers no refraction, that is, it passes undeviated. This is because, when it passes through the optical center, it is perpendicular to the curved surface of the lens. Hence, the angle of incidence is zero and therefore the angle of refraction is also zero.



54. Given, $m = -1$, negative sign means the image is real and inverted.

$\therefore |m| = 1$, it means that $h_i = h_o$ and this happens in convex lens only.

It implies that the object will be at $2F_1$ and the image will be at $2F_2$.

So, $v = u$.

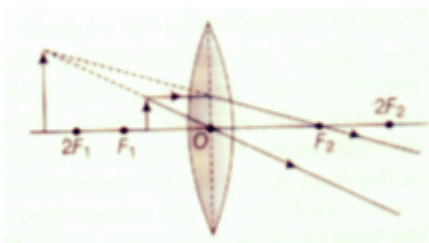
Object distance, $u = -30 \text{ cm}$

Image distance, $v = +30 \text{ cm}$

$\therefore$ According to lens formula.

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u} \Rightarrow \frac{1}{f} = \frac{1}{30} - \left(\frac{-1}{30} \right)$$

$$\Rightarrow f = 15 \text{ cm}$$



Now the object is moved 20 cm towards the lens, now the object will be at 10cm from

the lens and lie between F_1 and O.

$\therefore$ Object distance, $u = -10$ cm

Focal length, $f = +15$ cm

Again using lens formula,

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{v} = \frac{1}{15} + \frac{1}{(-10)}$$

$$\Rightarrow \frac{1}{v} = \frac{1}{15} - \frac{1}{10} = \frac{2-3}{30}$$

$$\Rightarrow v = -30 \text{ cm}$$

Hence, the image formed will be virtual, erect and highly magnified

55. Let Focal length, $f = 20$ cm

Object height, $h = 5$ cm

image distance, $v = -15$ cm

image height, $h' = ?$

object distance, $u = ?$

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\frac{1}{-20} = \frac{1}{-15} - \frac{1}{u}$$

$$\frac{1}{u} = \frac{-1}{15} + \frac{1}{20}$$

$$\frac{1}{u} = \frac{-4+3}{60} = \frac{-1}{60}$$

$$u = -60 \text{ cm}$$

$$\frac{h'}{h} = \frac{v}{u}$$

$$\frac{h'}{5} = \frac{-15}{-60}$$

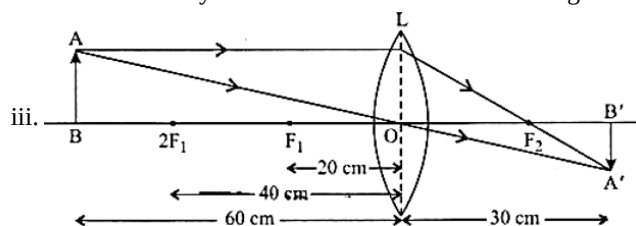
$$h' = \frac{15 \times 5}{60}$$

$$h' = \frac{5}{4} = +1.25 \text{ cm}$$

56. i. From the S. No. 3, we can say that the radius of curvature of the lens is 40 cm as distance of object and the distance of the image is same.

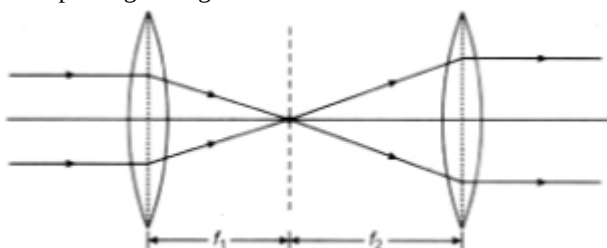
$$\therefore \text{Focal length, } f = \frac{R}{2} = \frac{40}{2} = 20 \text{ cm}$$

ii. S. No. 6 is not correct as for this observation the object distance is between focus and pole and for such cases, the image formed is always virtual but in this case a real image is forming as the image distance is positive.



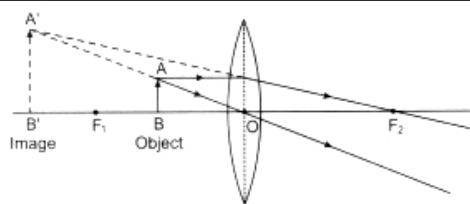
iv. Magnification $m = \frac{v}{u} = \frac{+30 \text{ cm}}{-60 \text{ cm}} = -0.5$

57. If the distance between two lenses become equal to sum of their focal lengths, then the parallel beam of light will emerge parallel after passing through the second lens.

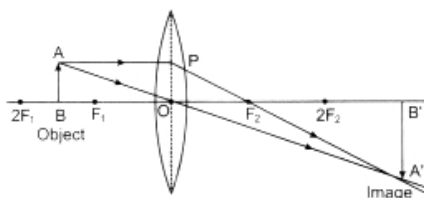


58.	Luminous	Non-luminous

The objects which possess light of its own are called luminous objects.	The object which does not possess light of its own, but receives light from external source and scatters it to the surroundings is called non-luminous.
Examples: The Sun, the stars, lamp, candle, etc.	Examples: The Moon, a chair, a book, etc.



59.



60. i. Visible spectrum is the band of coloured components of a white light beam.
 ii. Red light is scattered the least by air molecules and has longer wavelength. It travels the longest distance.
 iii. The given setup will behave like a glass slab, resulting in recombination of the seven colours to produce white light.

$$61. n = \frac{c}{v} = \frac{3 \times 10^8}{1.2 \times 10^8} = 2.5$$

62. i. The type of lens is convex because convex lenses have a positive power.

$$\text{ii. } P = \frac{1}{f}$$

$$+2 = \frac{1}{f}$$

$$f = +0.50 \text{ m}$$

Its focal length is +0.50 m.

$$\text{Using the relation } \frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\frac{1}{v} = \frac{1}{f} + \frac{1}{u}$$

$$= \frac{1}{0.5} + \frac{-1}{1}$$

$$= \frac{2+(-1)}{1}$$

distance of image $v = 1 \text{ m}$

63. Object height, $h_o = +10 \text{ cm}$

Focal length, $f = +15 \text{ cm}$

Object distance, $u = -25 \text{ cm}$

Image distance, $v = ?$

Image height, $h_i = ?$

Using lens formula,

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\frac{1}{15} = \frac{1}{v} - \frac{1}{(-25)}$$

$$\text{Or, } \frac{1}{v} = \frac{1}{15} - \frac{1}{25}$$

$$= \frac{1}{v} = \frac{2}{75}$$

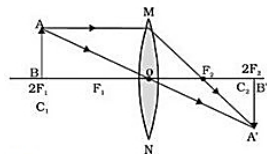
$$v = \frac{75}{2}$$

$$\text{Also, } \frac{-75}{-25} = \frac{1}{10}$$

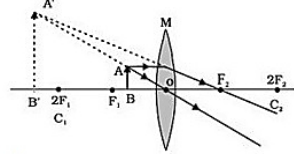
$$I = (-1.5) \times 10 = 15 \text{ cm}$$

Height of image formed is 15 cm, inverted.

64. A convex lens can form a (i) real, inverted and magnified image as well as (ii) virtual, erect and magnified image of an object

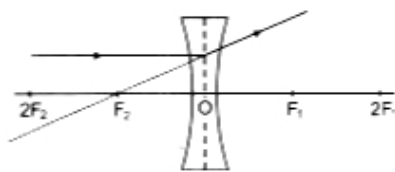


A) Object is between F and 2F

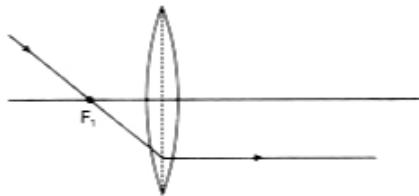


B) Object is between F and lens

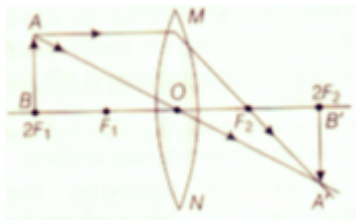
65. a.



b.



66. When object is placed at $2F_1$.



i. Image is formed at $2F_2$.

ii. Same size of image as that of the object.

iii. Real and inverted.

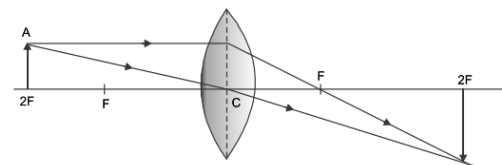
67. a. Same size of image is formed by a convex lens only when the object is placed at a distance $u=2f$ from the optical center and image is also formed at distance $2f$ on other side of lens.

1. The distance between the object and its image = 30cm

2. $2f = 30$

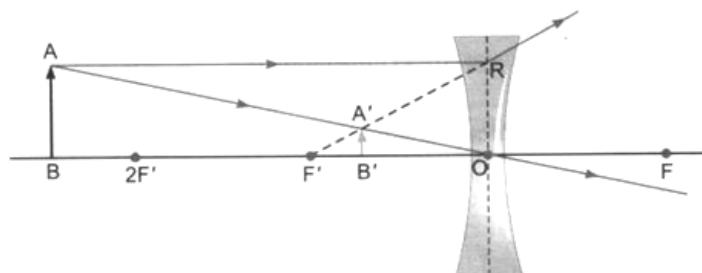
$f = 15\text{cm}$

b.



68. Since the lens is concave, hence f is negative

Given: $u = -30\text{ cm}$; $f = -10\text{ cm}$; $h = 2.5\text{ cm}$; $v = ?$; $h' = ?$



The lens formula for concave lens is $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

$$\frac{1}{v} - \frac{1}{-30} = \frac{1}{-10}$$

$$\frac{1}{v} = \frac{1}{10} - \frac{1}{30}$$

$$\frac{1}{v} = \frac{-3-1}{30}$$

$$v = -7.5\text{ cm}$$

The negative sign indicates the virtual nature of the image.

The image is at a distance of 7.5 cm from lens (in front of lens).

The magnification $m = \frac{v}{u} = \frac{-7.5}{30}$

$$= \frac{1}{4}$$

$$= +0.25$$

The positive sign with the magnification indicates that the image formed erect.

The size of the image is determined by h' .

$$\frac{h'}{h} = m$$

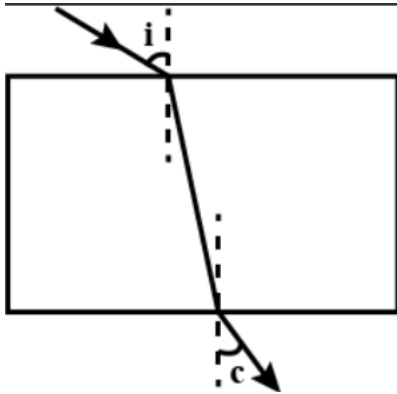
$$\begin{aligned}
 h' &= h \times m \\
 &= 2.5 \times 0.25 \\
 &= 0.625 \text{ m}
 \end{aligned}$$

Thus the image formed is virtual and erect. It is at a distance of 7.5 cm from lens and its size is 0.625 cm.

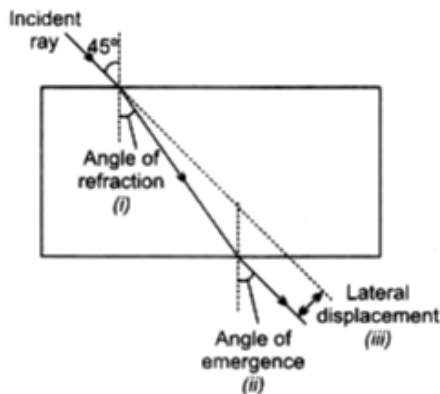
69. The two laws of refraction are:

- i. The incident ray, the refracted ray and the normal at the point of incidence, all lie in the same plane.
 - ii. The ratio of the sine of the angle of incidence (i) to the sine of angle of refraction (r) is a constant for the pair of given media.
- This constant is called the refractive index of the second medium with respect to the first medium.

70. When a rectangular glass slab is immersed in any medium, the ray incident on the slab suffers two refractions and emerges parallel to itself but displaced//shifted laterally as shown in the figure. This is because deviation suffered by the ray in the first refraction at one face of slab is equal and opposite to deviation suffered by the ray in 2nd refraction at the other face of the slab.



71.



72. As the image is obtained on the screen, it is real.

So,

Magnification, = -3

$v = 80 \text{ cm}$

$u = ?$

As $m = \frac{v}{u}$

so,

$$-3 = \frac{80}{u}$$

$$u = \frac{-80}{-3} \text{ cm}$$

From

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$= \frac{1}{80} + \frac{3}{80} = \frac{4}{80}$$

$$= \frac{1}{20}$$

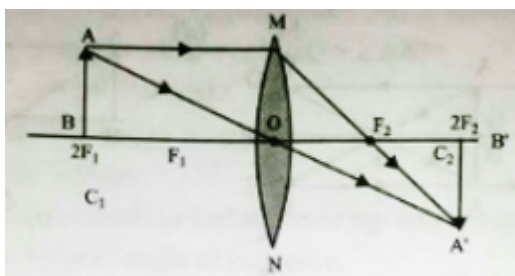
$$\frac{1}{f} = \frac{1}{20} \text{ cm}$$

so,

$f = 20 \text{ cm}$

The lens is convex and image formed at 80 cm from the lens is real and inverted.

73.



74. i. Light waves are refracted by some materials.
 ii. The image formed looks slightly bended because when light travels from air into water, it slows down, causing it to change direction slightly. This change of direction is called refraction. When light enters a more dense substance (higher refractive index), it 'bends' more towards the normal line.
 iii. The angle of refraction depends on the angle of incidence of the light, and the indexes of refraction of the two materials.
75. Power of lens is its ability to converge or diverge the light rays. A lens is small focal length converges or diverges the rays through a large angle and hence possess more power.

Power of lens in SI is measured by diopetre. Power of a lens is diopetre (D) is the reciprocal of focal length of lens in meters.

$$P = \frac{1}{f}$$

where P is in diopetre and f in meters. Powers are added algebraically. If lenses of power $P_1, P_2, P_3 \dots$ are placed in contact, their combined power is :

$$P = P_1 + P_2 + P_3 + \dots$$